Area Under Curve: Questions and Full Solutions

Question 01

English

The area of the region bounded by the curves $y = \log_2 x$, $y = \log_2(4 - x)$ and the x-axis is equal to

(A) $4 - \dfrac{1}{\log_e 2}$

(B) $2 - \dfrac{1}{\log_e 2}$

(C) $4 - \dfrac{2}{\log_e 2}$

(D) $\dfrac{2}{\log_e 2} - 2$

Only one option is correct.

हिन्दी

वक्रों $y = \log_2 x$, $y = \log_2(4 - x)$ तथा x-अक्ष द्वारा परिबद्ध क्षेत्र का क्षेत्रफल बराबर है

(A) $4 - \dfrac{1}{\log_e 2}$

(B) $2 - \dfrac{1}{\log_e 2}$

(C) $4 - \dfrac{2}{\log_e 2}$

(D) $\dfrac{2}{\log_e 2} - 2$

केवल एक विकल्प सही है।

Solution

Intersection of the curves: $\log_2 x = \log_2(4 - x) \Rightarrow x = 4 - x \Rightarrow x = 2$, $y = 1$. The first curve meets the x-axis at $x = 1$, the second at $x = 3$.

The bounded region lies above the x-axis, between $x = 1$ and $x = 3$, under the lower of the two curves. With vertical strips one would need $\int \log_2 x \, dx$ on two pieces; horizontal strips avoid this.

Solving each curve for x: $y = \log_2 x \Leftrightarrow x = 2^y$ and $y = \log_2(4 - x) \Leftrightarrow x = 4 - 2^y$.

For a fixed $y \in [0, 1]$ the horizontal segment inside the region runs from $x = 2^y$ to $x = 4 - 2^y$ (at $y = 1$ both equal 2), so its length is $4 - 2^{y+1} \geq 0$.

$$\text{Area} = \int_0^1 \left(4 - 2^{y+1}\right) dy = \left[4y - \frac{2^{y+1}}{\log_e 2}\right]_0^1 = 4 - \frac{4 - 2}{\log_e 2} = 4 - \frac{2}{\log_e 2}.$$

Check: $4 - \dfrac{2}{\log_e 2} \approx 1.115$, which is less than the area 2 of the triangle with vertices $(1, 0)$, $(3, 0)$, $(2, 1)$, as it must be since both curves are concave.

Correct Option: (C)

Question 02

English

Let C be the curve $\{(x, y) \in \mathbb{R}^2 : y^3 + 3y = 2x\}$, and let P be the point of C with abscissa 7. If O is the origin, then the area of the region bounded by C and the line segment OP is equal to

(A) 2
(B) 5
(C) 7
(D) 9

Only one option is correct.

हिन्दी

माना कि C वक्र $\{(x, y) \in \mathbb{R}^2 : y^3 + 3y = 2x\}$ है तथा P वक्र C का वह बिन्दु है जिसका भुज 7 है। यदि O मूलबिन्दु है, तो C तथा रेखाखण्ड OP द्वारा परिबद्ध क्षेत्र का क्षेत्रफल बराबर है

(A) 2
(B) 5
(C) 7
(D) 9

केवल एक विकल्प सही है।

Solution

C cannot be written easily as $y = f(x)$, but it is $x = g(y) = \dfrac{y^3 + 3y}{2}$ with $g'(y) = \dfrac{3y^2 + 3}{2} > 0$, so g is strictly increasing and C meets every vertical line exactly once.

For $x = 7$: $y^3 + 3y - 14 = 0 \Rightarrow (y - 2)(y^2 + 2y + 7) = 0$, and the quadratic has no real root, so $P = (7, 2)$.

The segment OP is $y = \dfrac{2x}{7}$, i.e. $x = \dfrac{7y}{2}$, $0 \le y \le 2$.

Between the two: $\dfrac{7y}{2} - g(y) = \dfrac{7y - y^3 - 3y}{2} = \dfrac{y(2 - y)(2 + y)}{2} \ge 0$ on $[0, 2]$, zero only at $y = 0, 2$. So C and OP meet only at O and P and bound one region.

Using horizontal strips: Area $= \displaystyle\int_0^2 \left(\frac{7y}{2} - \frac{y^3 + 3y}{2}\right) dy = \frac{1}{2}\int_0^2 (4y - y^3)\, dy = \frac{1}{2}\left[2y^2 - \frac{y^4}{4}\right]_0^2 = \frac{1}{2}(8 - 4) = 2.$

(Check: the area between C, the x-axis and $x = 7$ is $14 - \int_0^2 g(y)dy = 14 - 5 = 9$, the triangle under OP has area 7, and $9 - 7 = 2$.)

Correct Option: (A)

Question 03

English

The area of the region bounded by the curve $xy - 2x - 3y + 2 = 0$ and the line $x + y = 10$ is equal to

(A) $\frac{15}{2} - 4\log_e 2$

(B) $\frac{15}{2} - 8\log_e 2$

(C) $15 - 8\log_e 2$

(D) $\frac{9}{2} - 8\log_e 2$

Only one option is correct.

हिन्दी

वक्र $xy - 2x - 3y + 2 = 0$ तथा रेखा $x + y = 10$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल बराबर है

(A) $\frac{15}{2} - 4\log_e 2$

(B) $\frac{15}{2} - 8\log_e 2$

(C) $15 - 8\log_e 2$

(D) $\frac{9}{2} - 8\log_e 2$

केवल एक विकल्प सही है।

Solution

Group the terms: $xy - 2x - 3y + 6 = (x-3)(y-2)$, so $xy - 2x - 3y + 2 = (x-3)(y-2) - 4$ and the curve is $(x-3)(y-2) = 4$.

Shift the origin to $(3, 2)$ with $X = x - 3$, $Y = y - 2$. A translation does not change areas. The curve becomes $XY = 4$ (two branches, in the quadrants $X, Y > 0$ and $X, Y < 0$) and the line becomes $X + 3 + Y + 2 = 10$, i.e. $X + Y = 5$.

On the branch $X < 0$, $Y < 0$ we have $X + Y < 0 \neq 5$, so the line meets only the branch $X > 0$: $X(5 - X) = 4 \Rightarrow X^2 - 5X + 4 = 0 \Rightarrow X = 1,\ 4$, i.e. the points $(1, 4)$ and $(4, 1)$ in the new coordinates.

For $1 \le X \le 4$: $(5 - X) - \frac{4}{X} = \frac{-(X-1)(X-4)}{X} \ge 0$, so the line lies above the curve.

$$\text{Area} = \int_1^4 \left(5 - X - \frac{4}{X}\right) dX = \left[5X - \frac{X^2}{2} - 4\log_e X\right]_1^4 = 15 - \frac{15}{2} - 4\log_e 4 = \frac{15}{2} - 8\log_e 2.$$

Numerically this is about 1.955.

Correct Option: (B)

Question 04

English

The area of the region $\{(x, y) \in \mathbb{R}^2 : |x| + |y| \leq 2,\ |y| \geq x^2\}$ is equal to

(A) $\frac{10}{3}$

(B) $\frac{16}{3}$

(C) $\frac{13}{3}$

(D) $\frac{14}{3}$

Only one option is correct.

हिन्दी

क्षेत्र $\{(x, y) \in \mathbb{R}^2 : |x| + |y| \leq 2,\ |y| \geq x^2\}$ का क्षेत्रफल बराबर है

(A) $\frac{10}{3}$

(B) $\frac{16}{3}$

(C) $\frac{13}{3}$

(D) $\frac{14}{3}$

केवल एक विकल्प सही है।

Solution

Replacing x by $-x$ or y by $-y$ leaves both $|x| + |y| \leq 2$ and $|y| \geq x^2$ unchanged, so the region is symmetric about both axes and its area is four times the area of its part in $x \geq 0$, $y \geq 0$.

In the first quadrant the conditions are $y \leq 2 - x$ and $y \geq x^2$. Such y exists iff $x^2 \leq 2 - x$, i.e. $(x + 2)(x - 1) \leq 0$, so $0 \leq x \leq 1$. The parabola $y = x^2$ meets the side $x + y = 2$ of the square at $(1, 1)$.

$$\text{First-quadrant part} = \int_0^1 \left(2 - x - x^2\right) dx = \left[2x - \frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \frac{7}{6}.$$

$$\text{Area} = 4 \cdot \frac{7}{6} = \frac{14}{3}.$$

Check: the square has area 8; the removed part $|y| < x^2$ has first-quadrant area $\int_0^1 x^2 dx + \int_1^2 (2 - x) dx = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$, and $8 - 4 \cdot \frac{5}{6} = \frac{14}{3}$.

Correct Option: (D)

Question 05

English

The parabola $y = x^2$ and the line $y = 2x + 3$ meet at the points A and B. Let C be the point on the arc AB of the parabola at which the area of the triangle ABC is the greatest. Then the area of the part of the region bounded by the parabola and the line AB which lies outside the triangle ABC is equal to

(A) $\frac{8}{3}$

(B) $\frac{4}{3}$

(C) $\frac{32}{3}$

(D) $\frac{16}{3}$

Only one option is correct.

हिन्दी

परवलय $y = x^2$ तथा रेखा $y = 2x + 3$ बिन्दुओं A तथा B पर मिलते हैं। माना कि C परवलय के चाप AB पर वह बिन्दु है जिस पर त्रिभुज ABC का क्षेत्रफल अधिकतम है। तब परवलय तथा रेखा AB द्वारा परिबद्ध क्षेत्र के उस भाग का क्षेत्रफल, जो त्रिभुज ABC के बाहर है, बराबर है

(A) $\frac{8}{3}$

(B) $\frac{4}{3}$

(C) $\frac{32}{3}$

(D) $\frac{16}{3}$

केवल एक विकल्प सही है।

Solution

Intersections: $x^2 - 2x - 3 = 0 \Rightarrow x = -1, 3$, so $A(-1, 1)$ and $B(3, 9)$.

For $C(c, c^2)$ with $-1 < c < 3$, the area of ABC is $\frac{1}{2}|AB|$ times the distance of C from AB; the distance is proportional to $2c + 3 - c^2$, which is greatest at $c = 1$, where the tangent slope $2c$ equals the slope 2 of AB. So $C = (1, 1)$.

Region between the parabola and AB: Area $= \displaystyle\int_{-1}^{3} (2x + 3 - x^2)\, dx = \left[x^2 + 3x - \frac{x^3}{3}\right]_{-1}^{3} = 9 - \left(-\frac{5}{3}\right) = \frac{32}{3}$.

Triangle ABC: $\dfrac{1}{2}\,|(-1)(1-9) + 1(9-1) + 3(1-1)| = \dfrac{1}{2} \cdot 16 = 8$. This agrees with Archimedes' result that the segment is $\frac{4}{3}$ of this triangle (equivalently $\frac{2}{3}$ of the circumscribing parallelogram): $\frac{4}{3} \cdot 8 = \frac{32}{3}$.

The triangle lies inside the segment (its vertices are on the boundary of the convex segment), so the part outside the triangle has area $\dfrac{32}{3} - 8 = \dfrac{8}{3}$.

Check: this part is the two smaller segments on the chords AC ($y = 1$) and CB ($y = 4x - 3$), each of area $\dfrac{2^3}{6} = \dfrac{4}{3}$.

Correct Option: (A)

Question 06

English

For $m \in \mathbb{R}$, let $A(m)$ denote the area of the region bounded by the parabola $y = x^2$ and the line of slope m passing through the point $(1, 3)$. Let $A(m)$ attain its least value at $m = m_0$, and let $m_1 < m_2$ be the real numbers satisfying $A(m_1) = A(m_2) = \dfrac{32}{3}$. Then $(m_2 - m_0)(m_0 - m_1)$ is equal to

(A) 4
(B) 8
(C) 16
(D) $2\sqrt{2}$

Only one option is correct.

हिन्दी

$m \in \mathbb{R}$ के लिए, माना कि $A(m)$ परवलय $y = x^2$ तथा बिन्दु $(1, 3)$ से होकर जाने वाली ढाल m की रेखा द्वारा परिबद्ध क्षेत्र का क्षेत्रफल है। माना कि $A(m)$ अपना न्यूनतम मान $m = m_0$ पर प्राप्त करता है, तथा $m_1 < m_2$ वे वास्तविक संख्याएँ हैं जो $A(m_1) = A(m_2) = \dfrac{32}{3}$ को संतुष्ट करती हैं। तब $(m_2 - m_0)(m_0 - m_1)$ बराबर है

(A) 4
(B) 8
(C) 16
(D) $2\sqrt{2}$

केवल एक विकल्प सही है।

Solution

The line through $(1, 3)$ with slope m is $y = m(x - 1) + 3 = mx + 3 - m$. It meets $y = x^2$ where $x^2 - mx - (3 - m) = 0$.

Discriminant $D = m^2 + 4(3 - m) = (m - 2)^2 + 8 > 0$ for every m (the point $(1, 3)$ lies inside the parabola since $3 > 1^2$), so there are two intersection points with abscissae $\alpha < \beta$, $\beta - \alpha = \sqrt{D}$.

Between them the line is above the parabola and $mx + 3 - m - x^2 = (x - \alpha)(\beta - x)$. Hence $A(m) = \displaystyle\int_\alpha^\beta (x - \alpha)(\beta - x)\, dx = \frac{(\beta - \alpha)^3}{6} = \frac{D^{3/2}}{6}$ (substitute $x = \alpha + t$).

$A(m)$ is an increasing function of D, and $D = (m - 2)^2 + 8$ is least at $m = 2$. So $m_0 = 2$ and the least area is $\dfrac{8^{3/2}}{6} = \dfrac{8\sqrt{2}}{3}$.

$A(m) = \dfrac{32}{3} \Rightarrow D^{3/2} = 64 \Rightarrow D = 16 \Rightarrow (m - 2)^2 = 8 \Rightarrow m = 2 \pm 2\sqrt{2}$. So $m_1 = 2 - 2\sqrt{2}$, $m_2 = 2 + 2\sqrt{2}$.

$(m_2 - m_0)(m_0 - m_1) = (2\sqrt{2})(2\sqrt{2}) = 8$.

Correct Option: (B)

Question 07

English

Let a be a real number and let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that $\int_0^x f(t)\,dt = x^3 + ax^2$ for all $x \in \mathbb{R}$. The region bounded by the curve $y = f(x)$ and the x-axis lies in the fourth quadrant and has area 4. If $g : \mathbb{R} \to \mathbb{R}$ is defined by $g(x) = \int_0^x f(t)\,dt$, then the least value of g on $[0, \infty)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a एक वास्तविक संख्या है तथा $f : \mathbb{R} \to \mathbb{R}$ एक ऐसा सतत फलन है कि सभी $x \in \mathbb{R}$ के लिए $\int_0^x f(t)\,dt = x^3 + ax^2$ है। वक्र $y = f(x)$ तथा x-अक्ष द्वारा परिबद्ध क्षेत्र चतुर्थ चतुर्थांश में स्थित है तथा उसका क्षेत्रफल 4 है। यदि $g : \mathbb{R} \to \mathbb{R}$, $g(x) = \int_0^x f(t)\,dt$ द्वारा परिभाषित है, तो $[0, \infty)$ पर g का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

f is continuous, so by the first part of the fundamental theorem, $\frac{d}{dx}\int_0^x f(t)dt = f(x)$. Differentiating the identity gives $f(x) = 3x^2 + 2ax = x(3x + 2a)$.

$a \neq 0$, otherwise $f = 3x^2 \geq 0$ touches the axis only at 0 and bounds no region. The zeros are $x = 0$ and $x = -\frac{2a}{3}$, and $f < 0$ strictly between them (upward parabola).

So the bounded region lies below the x-axis between the two zeros. It is in the fourth quadrant iff these x are positive, i.e. $-\frac{2a}{3} > 0$, $a < 0$. (For $a > 0$ it would lie in the third quadrant.)

By the second part of the theorem, $\int_0^b f(t)dt = g(b) - g(0) = b^3 + ab^2$. With $b = -\frac{2a}{3}$: $\int_0^b f = b^2(b + a) = \frac{4a^2}{9} \cdot \frac{a}{3} = \frac{4a^3}{27}$, which is negative.

Area $= \left|\frac{4a^3}{27}\right| = -\frac{4a^3}{27} = 4 \Rightarrow a^3 = -27 \Rightarrow a = -3$. Then $f(x) = 3x(x - 2)$ and $g(x) = x^3 - 3x^2$.

On $[0, \infty)$: $g'(x) = f(x) < 0$ on $(0, 2)$ and > 0 on $(2, \infty)$, so the least value is $g(2) = 8 - 12 = -4$. It equals minus the area, because $g(2)$ is the signed integral of f over exactly the region below the axis.

Final Answer: -4

Question 08

English

Let $a > 0$ and let C be the curve $y = x^3 - a^2x$. The total area of the region enclosed between C and the x-axis from $x = 0$ to $x = 2a$ is 40. Then the area of the region bounded by C and the line $y = a^2x$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $a > 0$ है तथा C वक्र $y = x^3 - a^2x$ है। $x = 0$ से $x = 2a$ तक C तथा x-अक्ष के बीच घिरे क्षेत्र का कुल क्षेत्रफल 40 है। तब C तथा रेखा $y = a^2x$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$y = x^3 - a^2x = x(x - a)(x + a)$. For $0 < x < a$ it is negative and for $a < x < 2a$ it is positive, so the area between C and the x-axis over $[0, 2a]$ must be split at $x = a$ (the signed integral $\int_0^{2a} = 2a^4$ would be wrong).

With $F(x) = \dfrac{x^4}{4} - \dfrac{a^2x^2}{2}$: $F(0) = 0$, $F(a) = -\dfrac{a^4}{4}$, $F(2a) = 4a^4 - 2a^4 = 2a^4$.

Area $= |F(a) - F(0)| + |F(2a) - F(a)| = \dfrac{a^4}{4} + \dfrac{9a^4}{4} = \dfrac{5a^4}{2}$. Given $\dfrac{5a^4}{2} = 40$: $a^4 = 16$, and $a > 0$ gives $a = 2$.

C meets $y = a^2x$ where $x^3 - a^2x = a^2x$, i.e. $x(x^2 - 2a^2) = 0$: $x = 0, \pm\sqrt{2}\,a$. There are two bounded regions, over $[-\sqrt{2}a, 0]$ and $[0, \sqrt{2}a]$.

Both $x^3 - a^2x$ and a^2x are odd functions, so the figure is symmetric about the origin and the two regions have equal area.

On $(0, \sqrt{2}a)$: $a^2x - (x^3 - a^2x) = 2a^2x - x^3 > 0$. One region: $\displaystyle\int_0^{\sqrt{2}a} (2a^2x - x^3)dx = a^2 \cdot 2a^2 - \frac{4a^4}{4} = a^4$.

Total $= 2a^4 = 32$.

Final Answer: 32

Question 09

English

Consider the curves $y = \sin^{-1} x$ and $y = \cos^{-1} x$, $0 \le x \le 1$, where the inverse trigonometric functions take their principal values. Let A_1 be the area of the region bounded by these two curves and the line $x = 1$, and let A_2 be the area of the region bounded by these two curves and the y-axis. Then the value of $A_1 - A_2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वक्रों $y = \sin^{-1} x$ तथा $y = \cos^{-1} x$, $0 \le x \le 1$, पर विचार कीजिए, जहाँ प्रतिलोम त्रिकोणमितीय फलन अपने मुख्य मान लेते हैं। माना कि A_1 इन दोनों वक्रों तथा रेखा $x = 1$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल है, तथा A_2 इन दोनों वक्रों तथा y-अक्ष द्वारा परिबद्ध क्षेत्र का क्षेत्रफल है। तब $A_1 - A_2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

On $[0, 1]$, $\sin^{-1} x$ increases from 0 to $\dfrac{\pi}{2}$ and $\cos^{-1} x$ decreases from $\dfrac{\pi}{2}$ to 0. They are equal where $\sin^{-1} x = \dfrac{\pi}{4}$ (since the sum is $\dfrac{\pi}{2}$), i.e. at $P\left(\dfrac{1}{\sqrt{2}}, \dfrac{\pi}{4}\right)$.

So the difference $\cos^{-1} x - \sin^{-1} x$ changes sign at $x = \dfrac{1}{\sqrt{2}}$: the region with the y-axis lies over $\left[0, \dfrac{1}{\sqrt{2}}\right]$ and the region with $x = 1$ over $\left[\dfrac{1}{\sqrt{2}}, 1\right]$.

Vertical strips would need $\int \sin^{-1} x \, dx$; horizontal strips are simpler. Inverting: $y = \sin^{-1} x \Leftrightarrow x = \sin y$ and $y = \cos^{-1} x \Leftrightarrow x = \cos y$, $0 \le y \le \dfrac{\pi}{2}$.

Region A_2 (left of both curves, right of the y-axis): for $0 \le y \le \frac{\pi}{4}$ its horizontal strip is $0 \le x \le \sin y$; for $\frac{\pi}{4} \le y \le \frac{\pi}{2}$ it is $0 \le x \le \cos y$. $A_2 = [-\cos y]_0^{\pi/4} + [\sin y]_{\pi/4}^{\pi/2} = \left(1 - \dfrac{1}{\sqrt{2}}\right) + \left(1 - \dfrac{1}{\sqrt{2}}\right) = 2 - \sqrt{2}$.

Region A_1 (right of both curves, left of $x = 1$): for $0 \le y \le \frac{\pi}{4}$ the strip is $\cos y \le x \le 1$; for $\frac{\pi}{4} \le y \le \frac{\pi}{2}$ it is $\sin y \le x \le 1$. $A_1 = \left(\dfrac{\pi}{4} - \dfrac{1}{\sqrt{2}}\right) + \left(\dfrac{\pi}{4} - \dfrac{1}{\sqrt{2}}\right) = \dfrac{\pi}{2} - \sqrt{2}$.

$A_1 - A_2 = \dfrac{\pi}{2} - 2 \approx 1.5708 - 2 = -0.4292$, i.e. -0.43.

Check by the sign change: $A_1 - A_2 = \displaystyle\int_0^1 (\sin^{-1} x - \cos^{-1} x)\, dx$, and $\int_0^1 \sin^{-1} x \, dx = \dfrac{\pi}{2} - 1$, $\int_0^1 \cos^{-1} x \, dx = 1$, which gives $\dfrac{\pi}{2} - 2$ again.

Final Answer: -0.43

Question 10

English

The area of the region bounded by the curve $y = \left|x^2 - 2\right|$ and the line $y = x$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वक्र $y = \left|x^2 - 2\right|$ तथा रेखा $y = x$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Since $|x^2 - 2| \geq 0$, the line $y = x$ can meet the curve only for $x \geq 0$; for $x < 0$ the line lies below the curve and no region is formed there.

For $0 \leq x \leq \sqrt{2}$: $|x^2 - 2| = 2 - x^2$, and $x = 2 - x^2 \Rightarrow x^2 + x - 2 = 0 \Rightarrow x = 1$ (the root -2 is rejected).

For $x \geq \sqrt{2}$: $|x^2 - 2| = x^2 - 2$, and $x = x^2 - 2 \Rightarrow x = 2$ (the root -1 is rejected).

At $x = \sqrt{2}$ the curve is $0 < \sqrt{2}$, so the line lies above the curve on $(1, 2)$ and the bounded region is over $[1, 2]$.

$$\int_1^{\sqrt{2}} \left(x - (2 - x^2)\right) dx = \left[\frac{x^2}{2} + \frac{x^3}{3} - 2x\right]_1^{\sqrt{2}} = \left(1 - \frac{4\sqrt{2}}{3}\right) - \left(-\frac{7}{6}\right) = \frac{13}{6} - \frac{4\sqrt{2}}{3}.$$

$$\int_{\sqrt{2}}^2 \left(x - (x^2 - 2)\right) dx = \left[\frac{x^2}{2} - \frac{x^3}{3} + 2x\right]_{\sqrt{2}}^2 = \frac{10}{3} - \left(1 + \frac{4\sqrt{2}}{3}\right) = \frac{7}{3} - \frac{4\sqrt{2}}{3}.$$

$$\text{Area} = \frac{13}{6} + \frac{14}{6} - \frac{8\sqrt{2}}{3} = \frac{9}{2} - \frac{8\sqrt{2}}{3} \approx 0.7288, \text{ i.e. } 0.73.$$

Final Answer: 0.73

Question 11

English

The area of the region $\{(x, y) \in \mathbb{R}^2 : x^2 - 2|x| \leq y \leq 2|x| + 12\}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

क्षेत्र $\{(x, y) \in \mathbb{R}^2 : x^2 - 2|x| \leq y \leq 2|x| + 12\}$ का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Replacing x by $-x$ leaves both inequalities unchanged, so the region is symmetric about the y-axis. Its area is twice the area of the part with $x \geq 0$.

For $x \geq 0$ the region is $x^2 - 2x \leq y \leq 2x + 12$. A vertical segment exists iff $x^2 - 2x \leq 2x + 12$, i.e. $x^2 - 4x - 12 \leq 0$, $(x - 6)(x + 2) \leq 0$; with $x \geq 0$ this is $0 \leq x \leq 6$.

Its length is $(2x + 12) - (x^2 - 2x) = 12 + 4x - x^2$.

$$\int_0^6 (12 + 4x - x^2)dx = \left[12x + 2x^2 - \frac{x^3}{3}\right]_0^6 = 72 + 72 - 72 = 72.$$

Area $= 2 \cdot 72 = 144$.

Final Answer: 144

Question 12

English

Let $R = \{(x, y) \in \mathbb{R}^2 : |x| + |y| \leq 4 \leq |x| + |y| + |xy|\}$. Then the area of R is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $R = \{(x, y) \in \mathbb{R}^2 : |x| + |y| \leq 4 \leq |x| + |y| + |xy|\}$ है। तब R का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Each condition involves only $|x|$, $|y|$ and $|xy| = |x||y|$, so R is symmetric about both axes and its area is four times the area of $R \cap \{x \geq 0, y \geq 0\}$.

In the first quadrant: $x + y \leq 4$ (inside the square $|x| + |y| = 4$) and $x + y + xy \geq 4$. Adding 1: $1 + x + y + xy \geq 5$, i.e. $(1 + x)(1 + y) \geq 5$, so $y \geq \dfrac{5}{1 + x} - 1 = \dfrac{4 - x}{1 + x}$.

For $0 \leq x \leq 4$: $\dfrac{4 - x}{1 + x} \leq 4 - x$ because $4 - x \geq 0$ and $1 + x \geq 1$; equality only at $x = 0$ and $x = 4$.

So the first-quadrant part is $\dfrac{4 - x}{1 + x} \leq y \leq 4 - x$, $0 \leq x \leq 4$ (a region between the line and an arc of a rectangular hyperbola).

$$\text{Part} = \int_0^4 \left[(4 - x) - \left(\frac{5}{1 + x} - 1\right)\right] dx = \int_0^4 \left(5 - x - \frac{5}{1 + x}\right) dx = 20 - 8 - 5\log_e 5 = 12 - 5\log_e 5.$$

Area of $R = 4(12 - 5\log_e 5) = 48 - 20\log_e 5$. With $\log_e 5 \approx 1.609438$: $48 - 32.18876 = 15.81124$, i.e. 15.81.

Final Answer: 15.81

Question 13

English

Let $R = \{(x, y) \in \mathbb{R}^2 : y \geq x^2 - 2,\ x \geq y^2 - 2\}$. Then the area of R is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $R = \{(x, y) \in \mathbb{R}^2 : y \geq x^2 - 2,\ x \geq y^2 - 2\}$ है। तब R का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

If $(x, y) \in R$ then $(y, x) \in R$, because the two conditions are exchanged. So R is symmetric about the line $y = x$ and its area is twice the area of the part with $y \leq x$.

Corners of R: $y = x^2 - 2$ and $x = y^2 - 2$ give $y - x = x^2 - y^2$, i.e. $(y - x)(x + y + 1) = 0$.

On $y = x$: $x^2 - x - 2 = 0$, $x = 2$ or $x = -1$: the points $(2, 2)$ and $(-1, -1)$. On $x + y = -1$: $x^2 + x - 1 = 0$, $x = \beta = \dfrac{\sqrt{5} - 1}{2}$ or $x = -1 - \beta$: the points $(\beta, -1 - \beta)$ and $(-1 - \beta, \beta)$, mirror images in $y = x$.

Part below $y = x$. The condition $y \geq x^2 - 2$ with $y \leq x$ gives the parabolic segment $x^2 - 2 \leq y \leq x$, $-1 \leq x \leq 2$, of area $\displaystyle\int_{-1}^{2} (x + 2 - x^2)dx = \frac{(2 - (-1))^3}{6} = \frac{9}{2}$.

From it we must remove the points with $x < y^2 - 2$. For $y < 0$ this means $y < -\sqrt{x + 2}$. The lower parabola $y = x^2 - 2$ lies below $y = -\sqrt{x + 2}$ exactly for $-1 < x < \beta$ (they meet at $x = -1$ and at the corner $x = \beta$).

Removed area $r = \displaystyle\int_{-1}^{\beta} \left(-\sqrt{x + 2} - (x^2 - 2)\right) dx = \left[2x - \frac{x^3}{3} - \frac{2}{3}(x + 2)^{3/2}\right]_{-1}^{\beta}$.

With $\beta^2 = 1 - \beta$: $\beta^3 = \beta - \beta^2 = 2\beta - 1$, and $\beta + 2 = (\beta + 1)^2$ (since $(\beta + 1)^2 = \beta^2 + 2\beta + 1 = \beta + 2$), so $(\beta + 2)^{3/2} = (\beta + 1)^3 = \beta^3 + 3\beta^2 + 3\beta + 1 = 2\beta + 3$.

Value at β: $2\beta - \dfrac{2\beta - 1}{3} - \dfrac{2(2\beta + 3)}{3} = \dfrac{6\beta - 2\beta + 1 - 4\beta - 6}{3} = -\dfrac{5}{3}$. Value at -1: $-2 + \dfrac{1}{3} - \dfrac{2}{3} = -\dfrac{7}{3}$.

So $r = -\dfrac{5}{3} + \dfrac{7}{3} = \dfrac{2}{3}$.

Area of $R = 2\left(\dfrac{9}{2} - \dfrac{2}{3}\right) = \dfrac{23}{3} \approx 7.67$.

Final Answer: 7.67

Question 14

English

Let k be a real number and let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^3 + 3x^2 + kx + 1$. If the area of the region bounded by the curves $y = f(x)$ and $y = f(-x)$ is $\frac{9}{25}$, then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि k एक वास्तविक संख्या है तथा $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3 + 3x^2 + kx + 1$ द्वारा परिभाषित है। यदि वक्रों $y = f(x)$ तथा $y = f(-x)$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल $\frac{9}{25}$ है, तो k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$f(-x) = -x^3 + 3x^2 - kx + 1$, so $f(x) - f(-x) = 2x^3 + 2kx = 2x(x^2 + k)$: twice the odd part of f; the even part $3x^2 + 1$ cancels.

The curves meet where $x(x^2 + k) = 0$. If $k \geq 0$ the only meeting point is $x = 0$ and the difference keeps one sign on each side, so no bounded region exists. Hence $k < 0$, and the meeting points are $x = 0$, $x = \pm\sqrt{-k}$.

$y = f(-x)$ is the reflection of $y = f(x)$ in the y-axis, so the figure is symmetric about the y-axis; analytically, $|f(x) - f(-x)|$ is an even function, so $\int_{-c}^{c} |f(x) - f(-x)|dx = 2\int_{0}^{c} |f(x) - f(-x)|dx$.

On $(0, \sqrt{-k})$: $x^2 + k < 0$, so $|f(x) - f(-x)| = -2x(x^2 + k) = 2(-kx - x^3)$.

$$\text{Area} = 2\int_0^{\sqrt{-k}} 2(-kx - x^3)dx = 4\left[-\frac{kx^2}{2} - \frac{x^4}{4}\right]_0^{\sqrt{-k}} = 4\left(\frac{k^2}{2} - \frac{k^2}{4}\right) = k^2.$$

$k^2 = \frac{9}{25}$ with $k < 0$ gives $k = -\frac{3}{5} = -0.6$.

Final Answer: -0.6

Question 15

English

The area of the region bounded by the curve $y = x^3 - 6x^2 + 9x - 2$ and the line $y = 7x - 14$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वक्र $y = x^3 - 6x^2 + 9x - 2$ तथा रेखा $y = 7x - 14$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$y' = 3x^2 - 12x + 9$, $y'' = 6x - 12$, which vanishes at $x = 2$; $y(2) = 8 - 24 + 18 - 2 = 0$. So $(2, 0)$ is the point of inflection, and $7 \cdot 2 - 14 = 0$ shows that the line passes through it.

Shift the origin to $(2, 0)$: $x = X + 2$, $y = Y$. Areas are unchanged by a translation.

$(X+2)^3 - 6(X+2)^2 + 9(X+2) - 2 = X^3 + 6X^2 + 12X + 8 - 6X^2 - 24X - 24 + 9X + 18 - 2 = X^3 - 3X$.

The line becomes $Y = 7(X + 2) - 14 = 7X$.

Intersections: $X^3 - 3X = 7X \Rightarrow X(X^2 - 10) = 0 \Rightarrow X = 0, \pm\sqrt{10}$.

$X^3 - 3X - 7X = X^3 - 10X$ is odd, so the two regions (over $[-\sqrt{10}, 0]$ and $[0, \sqrt{10}]$) are symmetric about the new origin and have equal areas. On $(0, \sqrt{10})$ the line is above: $10X - X^3 > 0$.

$$\text{Area} = 2\int_0^{\sqrt{10}} (10X - X^3)dX = 2\left[5X^2 - \frac{X^4}{4}\right]_0^{\sqrt{10}} = 2(50 - 25) = 50.$$

Final Answer: 50

Question 16

English

For $k > 0$, let $f_k : \mathbb{R} \to \mathbb{R}$ be defined by $f_k(x) = \dfrac{x^3 + 3kx}{4k}$. If the area of the region bounded by the graphs of f_k and its inverse function f_k^{-1} is 50, then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$k > 0$ के लिए, माना कि $f_k : \mathbb{R} \to \mathbb{R}$, $f_k(x) = \dfrac{x^3 + 3kx}{4k}$ द्वारा परिभाषित है। यदि f_k तथा उसके प्रतिलोम फलन f_k^{-1} के आलेखों द्वारा परिबद्ध क्षेत्र का क्षेत्रफल 50 है, तो k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$f_k'(x) = \dfrac{3(x^2 + k)}{4k} > 0$ for all x, and $f_k(x) \to \pm\infty$ as $x \to \pm\infty$, so f_k is a strictly increasing bijection of $\mathbb{R}$ and f_k^{-1} exists.

For an increasing function, $f_k(x) = f_k^{-1}(x)$ implies $f_k(f_k(x)) = x$, which forces $f_k(x) = x$ (if $f_k(x) > x$ then $f_k(f_k(x)) > f_k(x) > x$, and similarly for $<$). So the two graphs meet only on $y = x$, at $f_k(x) = x$: $x^3 + 3kx = 4kx \Rightarrow x(x^2 - k) = 0 \Rightarrow x = 0, \pm\sqrt{k}$.

The graph of f_k^{-1} is the mirror image of that of f_k in $y = x$, so the area between f_k and f_k^{-1} is twice the area between f_k and $y = x$.

$x - f_k(x) = \dfrac{4kx - x^3 - 3kx}{4k} = \dfrac{x(k - x^2)}{4k}$, positive on $(0, \sqrt{k})$. Since f_k is odd, the piece over

$(-\sqrt{k}, 0)$ has the same area.

Area between f_k and $y = x$: $2\displaystyle\int_0^{\sqrt{k}} \frac{kx - x^3}{4k} dx = \frac{2}{4k}\left(\frac{k^2}{2} - \frac{k^2}{4}\right) = \frac{k}{8}$.

Area between f_k and $f_k^{-1} = \dfrac{k}{4}$. Setting $\dfrac{k}{4} = 50$ gives $k = 200$.

Final Answer: 200

Question 17

English

Let P be a point on the parabola $x^2 = -3y$. The tangent to this parabola at P meets the parabola $y = x^2$ at the points Q and R. If the area of the region bounded by the parabola $y = x^2$ and the line segment QR is $\dfrac{32}{3}$, then the ordinate of P is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि P परवलय $x^2 = -3y$ पर एक बिन्दु है। इस परवलय के बिन्दु P पर खींची गई स्पर्श रेखा परवलय $y = x^2$ को बिन्दुओं Q तथा R पर मिलती है। यदि परवलय $y = x^2$ तथा रेखाखण्ड QR द्वारा परिबद्ध क्षेत्र का क्षेत्रफल $\dfrac{32}{3}$ है, तो P की कोटि _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Any point of $x^2 = -3y$ is $P\left(a, -\dfrac{a^2}{3}\right)$. Differentiating, $2x = -3y'$, so the slope at P is $-\dfrac{2a}{3}$ and the tangent is $y + \dfrac{a^2}{3} = -\dfrac{2a}{3}(x - a)$, i.e. $y = -\dfrac{2a}{3}x + \dfrac{a^2}{3}$.

For $a = 0$ the tangent is $y = 0$, which touches $y = x^2$ at one point only, so $a \neq 0$.

Intersection with $y = x^2$: $x^2 + \dfrac{2a}{3}x - \dfrac{a^2}{3} = 0$. Discriminant $D = \dfrac{4a^2}{9} + \dfrac{4a^2}{3} = \dfrac{16a^2}{9} > 0$, so the roots $x_1 < x_2$ satisfy $x_2 - x_1 = \sqrt{D} = \dfrac{4|a|}{3}$.

Between Q and R the chord is above the parabola, and chord minus parabola $= (x - x_1)(x_2 - x)$.

So the area is $\displaystyle\int_{x_1}^{x_2} (x - x_1)(x_2 - x)dx = \frac{(x_2 - x_1)^3}{6} = \frac{1}{6} \cdot \frac{64|a|^3}{27} = \frac{32|a|^3}{81}$.

$\dfrac{32|a|^3}{81} = \dfrac{32}{3} \Rightarrow |a|^3 = 27 \Rightarrow a = \pm 3$.

In both cases the ordinate of P is $-\dfrac{a^2}{3} = -3$, so the answer is unique.

Final Answer: -3

Question 18

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 2x^2 - 2x$, and let $R = \left\{(x, y) \in \mathbb{R}^2 : |x| + |y| \leq 1,\ |y| \geq |f(|x|)|\right\}$. Then the area of R is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x^2 - 2x$ द्वारा परिभाषित है, तथा $R = \left\{(x, y) \in \mathbb{R}^2 : |x| + |y| \leq 1,\ |y| \geq |f(|x|)|\right\}$ है। तब R का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

R is described through $|x|$ and $|y|$ only, so it is symmetric about both axes and its area is four times the area of its part in the first quadrant.

Graph of $y = |f(|x|)|$: for $x \geq 0$, $f(x) = 2x(x-1) \leq 0$ on $[0, 1]$, so $|f(|x|)| = 2x - 2x^2$ there, a hump from $(0, 0)$ to $(1, 0)$ with greatest value $\dfrac{1}{2}$ at $x = \dfrac{1}{2}$; reflecting gives a second hump on $[-1, 0]$.

First quadrant: $y \geq 2x - 2x^2$ and $x + y \leq 1$. A vertical segment exists iff $2x - 2x^2 \leq 1 - x$, i.e. $2x^2 - 3x + 1 \geq 0$, $(2x - 1)(x - 1) \geq 0$. For $0 \leq x \leq 1$ this holds exactly when $x \leq \dfrac{1}{2}$ (or $x = 1$, a single point).

So the hump touches the side $x + y = 1$ at $\left(\dfrac{1}{2}, \dfrac{1}{2}\right)$ and for $\dfrac{1}{2} < x < 1$ lies above it; there is no part of R there.

$$\text{First-quadrant part} = \int_0^{1/2} \left[(1 - x) - (2x - 2x^2)\right] dx = \left[x - \frac{3x^2}{2} + \frac{2x^3}{3}\right]_0^{1/2} = \frac{1}{2} - \frac{3}{8} + \frac{1}{12} = \frac{5}{24}.$$

$$\text{Area of } R = 4 \cdot \frac{5}{24} = \frac{5}{6} = 0.8333\ldots, \text{ i.e. } 0.83.$$

Final Answer: 0.83

Question 19

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = [x]\{x\}$, where $[t]$ denotes the greatest integer less than or equal to t and $\{t\} = t - [t]$. For a real number $k < 0$, the area of the region bounded by the curve $y = f(x)$, the x-axis and the lines $x = k$ and $x = 4$ is $\frac{21}{2}$. Then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = [x]\{x\}$ द्वारा परिभाषित है, जहाँ $[t]$, t से छोटा या उसके बराबर महत्तम पूर्णांक है तथा $\{t\} = t - [t]$ है। किसी वास्तविक संख्या $k < 0$ के लिए, वक्र $y = f(x)$, x-अक्ष तथा रेखाओं $x = k$ एवं $x = 4$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल $\frac{21}{2}$ है। तब k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $n \le x < n+1$ (n an integer): $[x] = n$, $\{x\} = x - n$, so $f(x) = n(x-n)$, a straight segment rising (or falling) from 0 at $x = n$ towards n at $x = n+1$.

For $n > 0$ the piece is above the x-axis, for $n < 0$ below it, and for $n = 0$ it is on the axis. The area of each piece is therefore $\left|\int_n^{n+1} n(x-n)dx\right| = \frac{|n|}{2}$ (a triangle of base 1 and height $|n|$).

Over $[0, 4]$ the pieces $n = 0, 1, 2, 3$ give $\frac{0+1+2+3}{2} = 3$.

Over $[k, 0]$ with $k < 0$: each full piece $n = -1, -2, \ldots$ adds $\frac{|n|}{2}$; the area is a continuous, strictly increasing function of $|k|$ (every piece with $n \le -1$ has positive area).

For $k = -m$ (m a positive integer) the area over $[k, 0]$ is $\frac{1+2+\cdots+m}{2} = \frac{m(m+1)}{4}$. We need $\frac{21}{2} - 3 = \frac{15}{2}$, i.e. $m(m+1) = 30$, $m = 5$.

By strict monotonicity no other k works, so $k = -5$.

Final Answer: -5

Question 20

English

Let $R = \{(x, y) \in \mathbb{R}^2 : [x]^2 \leq y \leq 3 - 2[x]\}$, where $[t]$ denotes the greatest integer less than or equal to t. The line $x = \lambda$ divides R into two parts of equal area. Then the value of λ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $R = \{(x, y) \in \mathbb{R}^2 : [x]^2 \leq y \leq 3 - 2[x]\}$ है, जहाँ $[t]$, t से छोटा या उसके बराबर महत्तम पूर्णांक है। रेखा $x = \lambda$, R को समान क्षेत्रफल वाले दो भागों में विभाजित करती है। तब λ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Fix x with $[x] = n$, i.e. $n \leq x < n + 1$. The points of R above x are $n^2 \leq y \leq 3 - 2n$. This is non-empty iff $n^2 \leq 3 - 2n$, i.e. $n^2 + 2n - 3 \leq 0$, $(n + 3)(n - 1) \leq 0$, $-3 \leq n \leq 1$. So the x-values of R are $[-3, 2)$, exactly as for the domain of $\sqrt{3 - 2[x] - [x]^2}$.

The height of the section is $3 - 2n - n^2 = 4 - (n+1)^2$: for $n = -3$: 0; $n = -2$: 3; $n = -1$: 4; $n = 0$: 3; $n = 1$: 0.

For $n = -3$ and $n = 1$ the sections are single points ($y = 9$ and $y = 1$), giving segments of zero area. So R consists, up to area zero, of the rectangles $[-2, -1) \times [4, 7]$, $[-1, 0) \times [1, 5]$ and $[0, 1) \times [0, 3]$, with areas $3, 4, 3$; total 10.

The left area as a function of λ increases from 0 to 10 as λ goes from -2 to 1, strictly on $[-2, 1]$. After $[-2, -1)$ it is $3 < 5$; after $[-1, 0)$ it is $7 > 5$, so $-1 < \lambda < 0$ and $3 + 4(\lambda + 1) = 5$, $\lambda = -\dfrac{1}{2}$.

This agrees with the symmetry of the heights $3, 4, 3$ about the midpoint $x = -\dfrac{1}{2}$ of $[-2, 1]$.

Final Answer: -0.5

Question 21

English

For $t \in [1, 5]$, let $A(t)$ denote the area of the region bounded by the curve $y = x^3$, the tangent to this curve at the point (t, t^3), and the lines $x = 0$ and $x = 6$. If M and m are the greatest and the least values of $A(t)$ on $[1, 5]$, respectively, then the value of $M + m$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$t \in [1, 5]$ के लिए, माना कि $A(t)$ वक्र $y = x^3$, इस वक्र के बिन्दु (t, t^3) पर खींची गई स्पर्श रेखा, तथा रेखाओं $x = 0$ एवं $x = 6$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल है। यदि $[1, 5]$ पर $A(t)$ के महत्तम तथा न्यूनतम मान क्रमशः M तथा m हैं, तो $M + m$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$y' = 3x^2$, so the tangent at (t, t^3) is $y = t^3 + 3t^2(x - t) = 3t^2x - 2t^3$.

Curve minus tangent: $x^3 - 3t^2x + 2t^3 = (x - t)^2(x + 2t)$. For $t \in [1, 5]$ and $x \in [0, 6]$, $x + 2t > 0$, so the difference is ≥ 0, zero only at $x = t$: the curve lies above the tangent throughout the strip and the region is well defined.

$$A(t) = \int_0^6 \left(x^3 - 3t^2x + 2t^3\right) dx = \frac{6^4}{4} - \frac{3t^2 \cdot 36}{2} + 12t^3 = 324 - 54t^2 + 12t^3.$$

$A'(t) = -108t + 36t^2 = 36t(t - 3)$. On $[1, 5]$ the only critical point is $t = 3$; $A' < 0$ on $(1, 3)$ and $A' > 0$ on $(3, 5)$.

Least value: $m = A(3) = 324 - 486 + 324 = 162$ (the tangent at the midpoint of the strip).

Greatest value on the closed interval is at an end point: $A(1) = 324 - 54 + 12 = 282$, $A(5) = 324 - 1350 + 1500 = 474$. So $M = 474$.

$M + m = 474 + 162 = 636$.

Final Answer: 636